

Using the file `typemap.xml`, `polyconvert` imports polygons from OSM-data and produces a SUMO-polygon file.

`Polyconvert` imports geometrical shapes (polygons or points of interest) from different sources and converts them to a representation that may be visualised using SUMO-GUI.

```
polyconvert --net-file map.
net.xml --osm-files map.osm
--type-file typemap.xml -o
map.poly.xml
```

`randomTrips.py` generates a set of random trips for a given network (option `-n`).

It does so by choosing the source and destination edge either uniformly at random or weighted by length (option `-l`), by number of lanes (option `-L`) or both. The resulting trips are stored in an xml file (option `-o`, default `trips.xml`) suitable for the DUAROUTER, which is called automatically if the `-r` option (with a file name for the resulting route file) is given. The trips are distributed evenly in an interval defined by the begin (option `-b`, default `0`) and end time (option `-e`, default `3600`) in seconds. The number of trips is defined by the repetition rate (option `-p`, default `1`) in seconds. Every trip has an ID consisting of a prefix (option `--prefix`, default `"`) and a running number.

```
python /home/kunal/sumo-0.22.0/tools/trip/
randomTrips.py -n map.net.xml -e 100 -l
python /home/kunal/sumo-0.22.0/tools/trip/
randomTrips.py -n map.net.xml -r map.rou.xml -e
100 -l
```

Now, you need to make a `sumo.cfg` file for the configuration of the `sumo-gui`. This is what your `sumo.cfg` file looks like:

```
<configuration xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:noNamespaceSchemaLocation=
"http://sumo.sf.net/xsd/sumoConfiguration.xsd">

<input>
  <net-file value="map.net.xml"/>
  <route-files value="map.rou.xml"/>
  <additional-files value="map.poly.xml"/>
</input>
```

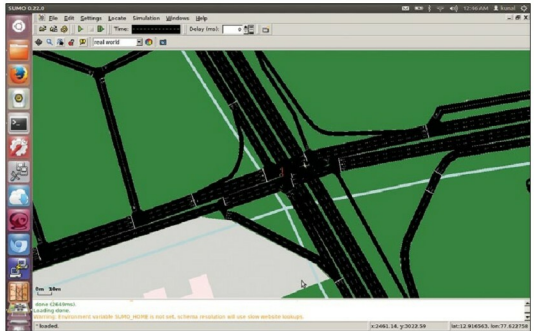


Figure 2: Final map after feeding command into SUMO

```
<time>
  <begin value="0"/>
  <end value="1000"/>
  <step-length value="0.1"/>
</time>
</configuration>
```

Save the above file with a file name. I saved it as `map.sumo.cfg`.

Now run the following command from the terminal:

```
sumo-gui map.sumo.cfg
```

After you enter the command, the SUMO-GUI window will pop up. Figure 2 shows what your window should look like.

If you are interested in vehicular ad hoc networks (VANETs), this article can be your starting point. It will teach you how to simulate real-time networks. You can then proceed further and add your research or project component to it.

You can refer to the complete video on YouTube;

here is the link: <https://www.youtube.com/watch?v=zm5h9H5OS8>



References

- [1] <http://www.dlr.de/ts/en/desktopdefault.aspx>

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The author is studying for an M. Tech in computer science and engineering at the VIT University in Chennai. His areas of interest are networking (including wireless networks) and open source technologies like NS2, SUMO and Linux. A gadget lover, he has recently started blogging on the latest devices and the technologies they use. He is the author of www.technerina.com